

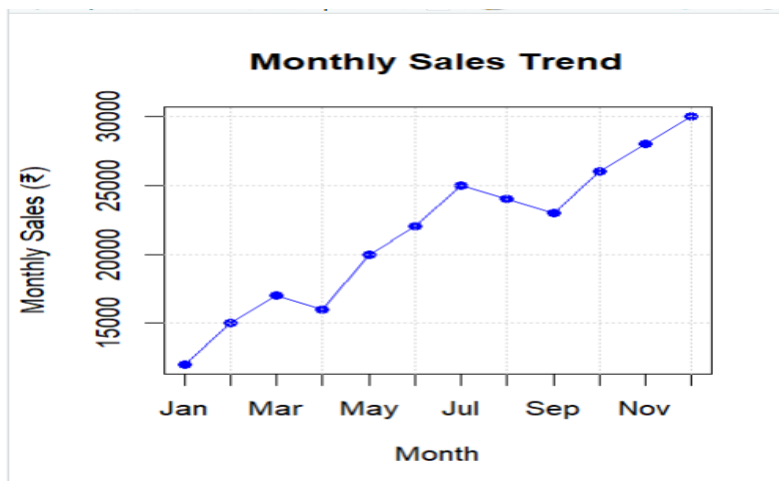
Set 1: Monthly Sales Data

1: Create a Line Chart to Visualize Monthly Sales

R Program

```
months <- c("Jan","Feb","Mar","Apr","May","Jun",  
            "Jul","Aug","Sep","Oct","Nov","Dec")  
sales <- c(12000,15000,17000,16000,20000,22000,  
           25000,24000,23000,26000,28000,30000)  
  
# Create Data Frame  
sales_data <- data.frame(Month=months, Sales=sales)  
  
# Line Chart  
plot(sales_data$Sales,  
      type="o",  
      col="blue",  
      pch=16,  
      xaxt="n",  
      xlab="Month",  
      ylab="Monthly Sales (₹)",  
      main="Monthly Sales Trend")  
  
axis(1, at=1:12, labels=sales_data$Month)  
grid()
```

Output



2: Generate a Bar Chart Showing Top-Selling Products

R Program

```
# Product Sales Data

products <- c("Laptop","Mobile","Tablet","Headphones","Smartwatch")

sales <- c(350,520,180,300,260)

# Bar Chart

barplot(sales,

        names.arg=products,

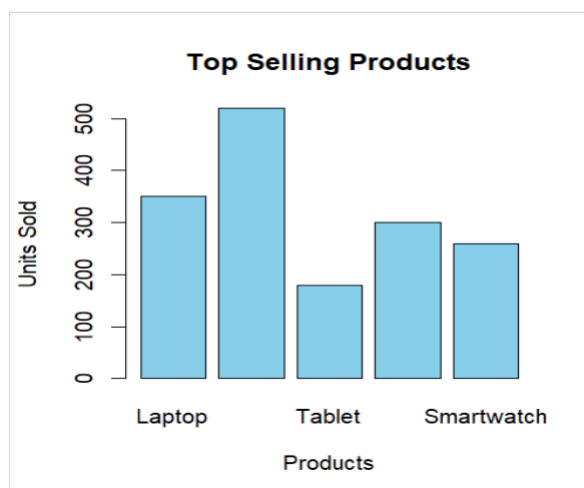
        col="skyblue",

        main="Top Selling Products",

        xlab="Products",

        ylab="Units Sold")
```

Output



3: Scatter Plot Between Advertising Budget and Monthly Sales

R Program

```
# Advertising Budget vs Monthly Sales

advertising <- c(10,12,15,14,18,20,22,21,20,24,26,28)

sales <- c(12000,15000,17000,16000,20000,22000,

          25000,24000,23000,26000,28000,30000)

plot(advertising,

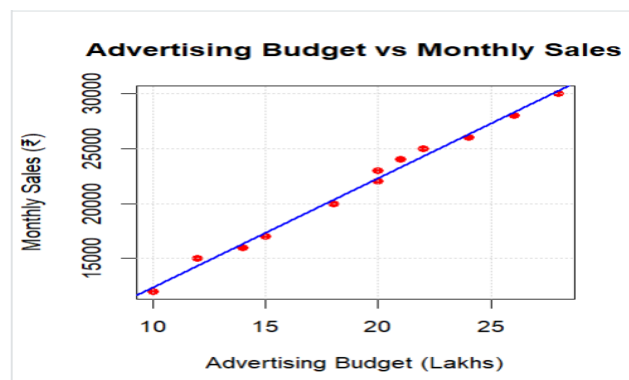
     sales,
```

```

pch=19,
col="red",
xlab="Advertising Budget (Lakhs)",
ylab="Monthly Sales (₹)",
main="Advertising Budget vs Monthly Sales")
grid()
# Add regression line
abline(lm(sales~advertising), col="blue", lwd=2)

```

Output



4: Interactive Dashboard (Line Chart + Bar Chart)

R Program

```

library(shiny)
ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      h4("Sales Dashboard")
    ),
    mainPanel(
      plotOutput("lineplot"),
      br(),
      plotOutput("barplot")
    )
  )
)

```

```

    )
  )
)
server <- function(input, output){
  output$lineplot <- renderPlot({
    months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
               "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
    sales <- c(12000, 15000, 17000, 16000, 20000, 22000,
               25000, 24000, 23000, 26000, 28000, 30000)
    plot(sales,
         type="o",
         col="blue",
         pch=16,
         xaxt="n",
         xlab="Month",
         ylab="Sales",
         main="Monthly Sales Trend")

    axis(1, at=1:12, labels=months)
  })
  output$barplot <- renderPlot({
    products <- c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch")
    productSales <- c(350, 520, 180, 300, 260)
    barplot(productSales,
            names.arg=products,
            col="orange",
            xlab="Products",
            ylab="Units Sold",
            main="Top Selling Products")
  })
}

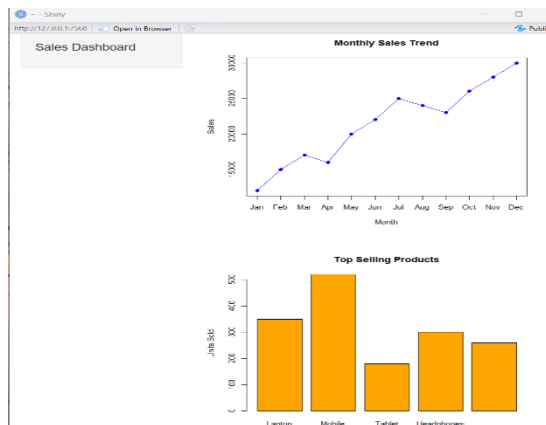
```

```

    })
  }
  shinyApp(ui=ui, server=server)

```

Output



Set 2: Customer Feedback Analysis

1: Create a Histogram for Customer Ages

R Program

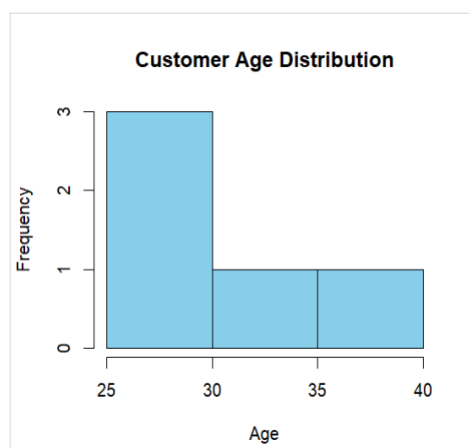
```
age <- c(25,30,35,28,40)
```

```

hist(age,
      col="skyblue",
      main="Customer Age Distribution",
      xlab="Age",
      ylab="Frequency")

```

Output



2. Pie Chart of Customer Satisfaction Scores

```
score <- c(4,5,3,4,5)
pie(table(score),
    labels=c("3 Stars","4 Stars","5 Stars"),
    col=c("pink","lightgreen","skyblue"),
    main="Customer Satisfaction")
```

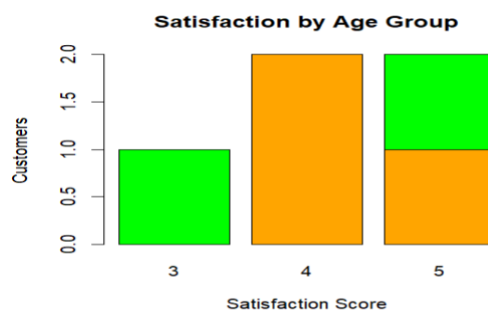
Output



3. Stacked Bar Chart

```
agegroup <- c("21-30","21-30","31-40","21-30","31-40")
score <- c(4,5,3,4,5)
data <- table(agegroup, score)
barplot(data,
    col=c("orange","green"),
    main="Satisfaction by Age Group",
    xlab="Satisfaction Score",
    ylab="Customers")
```

Output



4.Word Cloud

```
feedback <- c("Good","Excellent","Good","Service","Excellent","Quality")  
barplot(table(feedback),  
        col="lightblue",  
        main="Customer Feedback",  
        xlab="Words",  
        ylab="Frequency")
```

Output



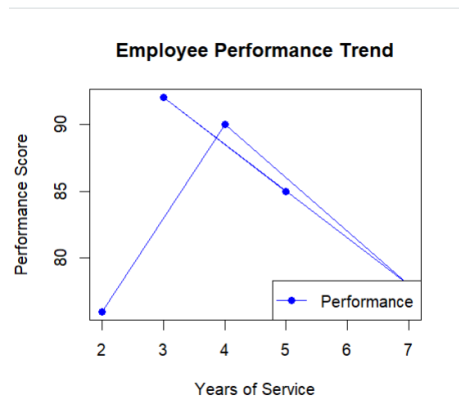
Set 3: Employee Performance Evaluation

1. Line Chart – Employee Performance Trend

```
service <- c(5,3,7,4,2)  
performance <- c(85,92,78,90,76)  
plot(service, performance,  
      type="o",  
      col="blue",  
      pch=16,  
      xlab="Years of Service",  
      ylab="Performance Score",  
      main="Employee Performance Trend")  
legend("bottomright",  
      legend="Performance",  
      col="blue",
```

```
lty=1,
pch=16)
```

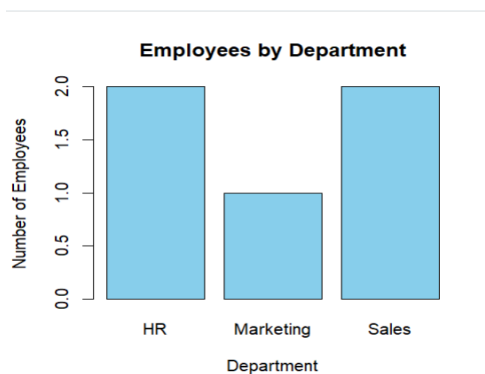
Output



2. Bar Chart – Employees by Department

```
department <- c("Sales","HR","Marketing","Sales","HR")
barplot(table(department),
        col="skyblue",
        main="Employees by Department",
        xlab="Department",
        ylab="Number of Employees")
```

Output



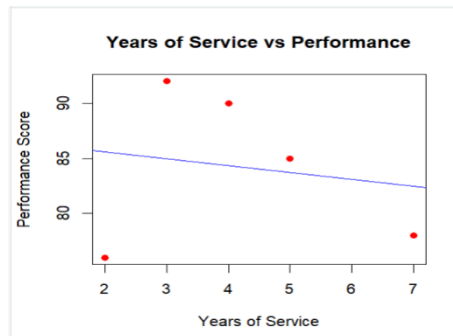
3. Scatter Plot – Years of Service vs Performance

```
service <- c(5,3,7,4,2)
performance <- c(85,92,78,90,76)
plot(service, performance,
      pch=19,
```



```
col="red",
xlab="Years of Service",
ylab="Performance Score",
main="Years of Service vs Performance")
abline(lm(performance~service), col="blue")
```

Output



4. Tableau Dashboard

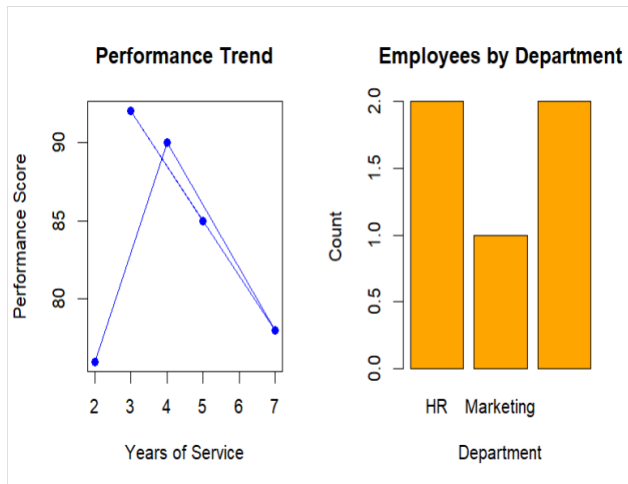
```
par(mfrow=c(1,2))
# Line Chart
service <- c(5,3,7,4,2)
performance <- c(85,92,78,90,76)
plot(service, performance,
      type="o",
      col="blue",
      pch=16,
      xlab="Years of Service",
      ylab="Performance Score",
      main="Performance Trend")
# Bar Chart
department <- c("Sales","HR","Marketing","Sales","HR")
barplot(table(department),
        col="orange",
        main="Employees by Department",
        xlab="Department",
```

```

        ylab="Count")
par(mfrow=c(1,1))

```

Output



SET 4: Product Inventory Management

Q1. Bar Chart

```

product <- c("A","B","C","D","E")
quantity <- c(250,175,300,200,220)
barplot(quantity,
        names.arg=product,
        col="skyblue",
        main="Product Inventory",
        xlab="Products",
        ylab="Quantity")

```

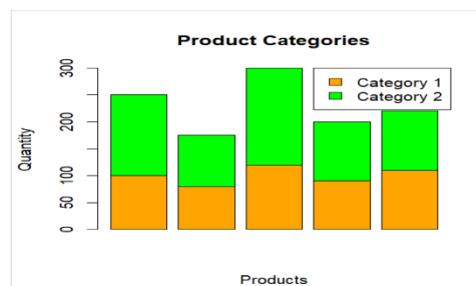
Output



Q2. Stacked Bar Chart

```
data <- matrix(c(100,80,120,90,110,  
                150,95,180,110,110),  
              nrow=2, byrow=TRUE)  
barplot(data,  
        col=c("orange","green"),  
        main="Product Categories",  
        xlab="Products",  
        ylab="Quantity")  
legend("topright",  
      legend=c("Category 1","Category 2"),  
      fill=c("orange","green"))
```

Output



Q3. Scatter Plot

```
price <- c(100,150,80,120,130)  
quantity <- c(250,175,300,200,220)  
plot(price, quantity,  
     pch=19,  
     col="red",  
     xlab="Price",  
     ylab="Quantity",  
     main="Price vs Quantity")  
abline(lm(quantity~price), col="blue")
```

Output



Q4. Tableau Dashboard

```
M par(mfrow=c(1,2))
```

```
# Bar Chart
```

```
product <- c("A","B","C","D","E")
```

```
quantity <- c(250,175,300,200,220)
```

```
barplot(quantity,
```

```
  names.arg=product,
```

```
  col="skyblue",
```

```
  main="Product Inventory")
```

```
# Stacked Bar Chart
```

```
data <- matrix(c(100,80,120,90,110,
```

```
  150,95,180,110,110),
```

```
  nrow=2,
```

```
  byrow=TRUE)
```

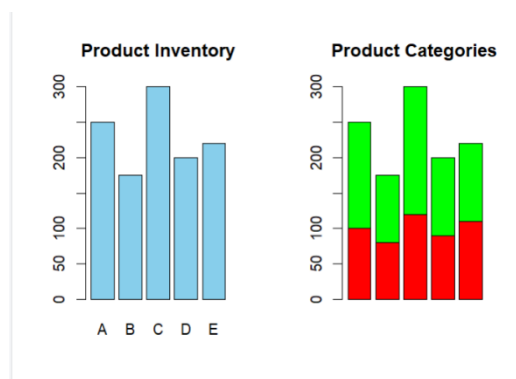
```
barplot(data,
```

```
  col=c("red","green"),
```

```
  main="Product Categories")
```

```
par(mfrow=c(1,1))
```

Output



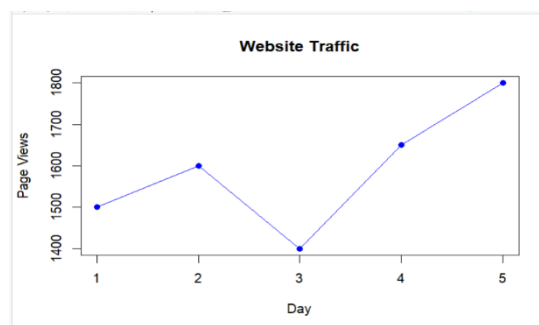
SET 5: Website Analytics

Q1. Line Chart

```
views <- c(1500,1600,1400,1650,1800)
```

```
plot(views,  
     type="o",  
     col="blue",  
     pch=16,  
     xlab="Day",  
     ylab="Page Views",  
     main="Website Traffic")
```

Output

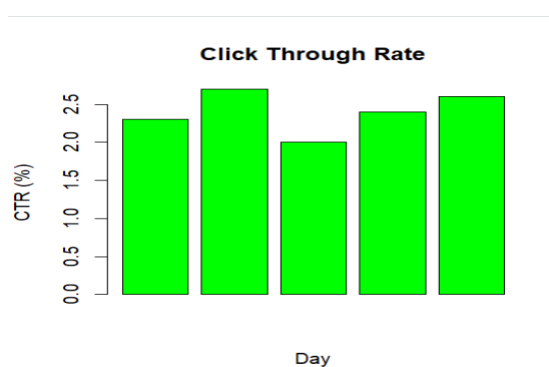


Q2. Bar Chart

```
ctr <- c(2.3,2.7,2.0,2.4,2.6)
```

```
barplot(ctr,  
        col="green",  
        main="Click Through Rate",  
        xlab="Day",  
        ylab="CTR (%)")
```

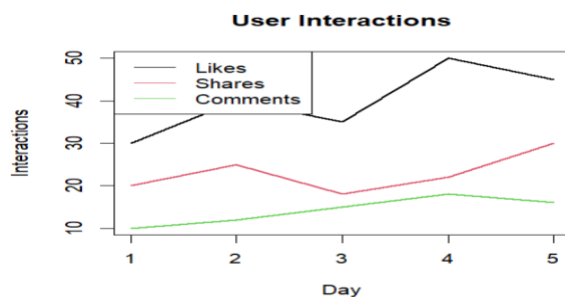
Output



Q3. Stacked Area Chart

```
likes <- c(30,40,35,50,45)
shares <- c(20,25,18,22,30)
comments <- c(10,12,15,18,16)
data <- rbind(likes,shares,comments)
matplot(t(data),
        type="l",
        lty=1,
        lwd=2,
        xlab="Day",
        ylab="Interactions",
        main="User Interactions")
legend("topleft",
       legend=c("Likes","Shares","Comments"),
       col=1:3,
       lty=1)
```

Output



Q4. Tableau Dashboard

```
par(mfrow=c(1,1))
```

```
par(mfrow=c(1,2))
```

```
# Line Chart
```

```
views <- c(1500,1600,1400,1650,1800)
```

```
plot(views,  
     type="o",  
     col="blue",  
     pch=16,  
     main="Website Traffic",  
     xlab="Day",  
     ylab="Page Views")
```

```
# Bar Chart
```

```
ctr <- c(2.3,2.7,2.0,2.4,2.6)
```

```
barplot(ctr,  
        col="green",  
        main="Click Through Rate",  
        xlab="Day",  
        ylab="CTR (%)")
```

```
par(mfrow=c(1,1))
```

Output

